Department of Artificial Intelligence,SVNIT
Subject- Machine Learning (AI301)
LAB ASSIGNMENT

Datasets: Synthetic NumPy data (Q1–Q4); UCI Wine Quality — red wine (winequality-red.csv, semicolon-delimited) (Q5–Q6); Iris (Q7)

Questions:

1. Generate X = 2*np.random.rand(100,1), y = 4 + 3*X + np.random.randn(100,1). Implement Batch Gradient Descent from scratch (learning rate = 0.1, 1000 iterations, random initial θ). Plot MSE vs. iteration number and report the final θ values.

2. Implement Stochastic Gradient Descent on the same data (one random instance per step, learning schedule η = t0/(t+t1) with t0=5, t1=50, 50 epochs). Overlay its MSE trajectory against Batch GD on one plot and report the total number of parameter updates each method performed.

3. Implement Mini-batch GD (batch_size=20, 50 epochs). Report the final θ for all three GD variants and the number of epochs each took to reach MSE < 1.0, in one table.

4. Generate y = 0.5*X**2 + X + 2 + np.random.randn(*X.shape) for X in [-3, 3]. Fit polynomial regression at degrees 1, 2, and 300 (PolynomialFeatures + LinearRegression). Plot all three fitted curves against the training data on one figure. Plot learning curves (train/validation RMSE vs. training-set size) for degree 1 and degree 300 to show underfitting vs. overfitting.

5. On winequality-red.csv, fit Ridge, Lasso, and ElasticNet predicting quality for alpha ∈ {0.001, 0.01, 0.1, 1, 10}. Report test RMSE for every (model, alpha) pair in one table. Plot Lasso's coefficient value vs. alpha (one line per feature) and report which feature's coefficient reaches zero first.

6. Binarize quality into good (≥7) vs. not good (<7). Fit LogisticRegression on an 80/20 split. Report accuracy, precision, recall, and F1-score.

7. On Iris, using only petal length and petal width, fit LogisticRegression(multi_class="multinomial"). Plot the resulting 3-class decision boundary as a filled contour plot with training points overlaid